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“Mixed-signal design is very different from VLSI”

Pericom Semiconductor enables serial connectivity with complete solutions for computing, communications and consumer market segments. With product range spanning from analogue, digital and mixed-signal integrated circuits, power management solutions to quartz-based frequency control products (FCP), Pericom supplies essential solutions for timing, switching, bridging and conditioning of high-speed signals.

Kay Annamalai, director of marketing, Pericom Semiconductor, spoke to Dilin Anand and Abhishek Mutha of EFY about the importance of clocks, common design challenges and the difficulties faced in mixed-signal design

Could you elaborate on the Internet of Things—growth of smart devices using technology like low-power Bluetooth 4.0—and how selecting the right components will help design engineers?

Today, all objects and humans are interconnected via Internet and other wireless connectivity options. Materials and vehicles are tagged and monitored, buildings are secured through surveillance, and energy storage and transmission are all monitored and controlled through smart meters that are connected to the network.

To promote this further, we need small, low-power devices for monitoring and connectivity. Components like low-power, small-size crystal oscillators, tiny load switches and level shifters will all help in designing these small devices. Overall, designers need to focus on reducing power consumption of these systems.

What are the latest solutions that help reduce power consumption in a circuit?

The latest power switches offer very low resistance, standby and shutdown current, and also protect against current-overload and short-circuit conditions in a circuit. On the other hand, the latest low dropout regulators (LDOs) make less noise and offer high ripple rejection and low

standby current. Also, current microprocessor supervisory circuits offer voltage-monitoring, power-up reset and watchdog functions. A combination of these solutions will enable engineers to build more efficient circuits that would consume much less power. For embedded designs, engineers can utilise microprocessor supervisory circuits, power switches and LDOs.

How important are clocks—be it ICs or real-time—for enabling high-speed applications?

Clocks are the heart of every system. Not only do they need to be stable and accurate but they also need to perform in terms of jitter, especially for high-speed applications like 10Gb Ethernet and 12Gb SAS in storage. Moreover, they need to consume little power.

What are the latest advances in timing solutions available for design engineers?

The latest timing solutions are noteworthy, as they now include low-jitter, high-frequency and small-sized crystal oscillators, as well as clock generators that integrate multiple frequencies and multiple copies of the frequencies. We also have high-speed signal-conditioning serial connectivity solutions up to 12.5 Gbps for high-speed backplanes, which